

Type # 1

(1) $\frac{1}{(3^{-1})^2} = ?$

- (A)
- $\frac{1}{3}$
- (B)
- $\frac{1}{9}$
- (C) 9 (D) 3

Solution :

$$\Rightarrow \frac{1}{3^{-1 \times 2}} = \frac{1}{3^{-2}} = 3^2 = 9$$

 $\therefore$ (C) 9

(2) $(-2)^3 \times (-2)^{-3} = ?$

- (A) 1 (B) 0 (C) 2 (D) 4

Solution :

$$\Rightarrow (-2)^{3+(-3)} = (-2)^0 = 1$$

 $\therefore$ (A) 1

(3) $x^5 \times x \div x^6 = ?$

- (A) 1 (B)
- x
- (C)
- x^{12}
- (D)
- x^6

Solution :

$$\begin{aligned} \Rightarrow x^5 \times x \times \frac{1}{x^6} &= \frac{x^{5+1}}{x^6} \\ &= \frac{x^6}{x^6} \\ &= x^{6-6} \\ &= x^0 \end{aligned}$$

 $\therefore$ (A) 1

(4) $\left[\left(\frac{1}{2}\right)^2\right]^{-2} = ?$

- (A) 4 (B) 8 (C) 16 (D) -16

Solution :

$$\begin{aligned} \Rightarrow \left(\frac{1}{2}\right)^{2 \times -2} &= \left(\frac{1}{2}\right)^{-4} \\ &= (2)^{+4} \\ &= 2 \times 2 \times 2 \times 2 \\ &= 16 \end{aligned}$$

 $\therefore$ (C) 16

(5) $(-12x^{-2}y^{-3})^0 = ?$

- (A) 0 (B) 12
-
- (C) 120 (D) 1

Solution :

 $\therefore$ (D) 1

(6) $3^{10} \div 3^{12} = ?$

- (A) 9 (B) 3 (C)
- $\frac{1}{9}$
- (D) -9

Solution :

$$\Rightarrow \frac{3^{10}}{3^{12}} = 3^{10-12} = 3^{-2}$$

$$\Rightarrow \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

 $\therefore$ (C) $\frac{1}{9}$

(7) $\left(\frac{4}{9}\right)^{-2} = ?$

- (A)
- $\frac{2}{3}$
- (B)
- $\frac{3}{2}$
- (C)
- $-\frac{2}{3}$
- (D)
- $\frac{81}{16}$

Solution :

$$\begin{aligned} \Rightarrow \left(\frac{4}{9}\right)^{-2} \\ \left(\frac{9}{4}\right)^2 &= \frac{81}{16} \end{aligned}$$

 $\therefore$ (D) $\frac{81}{16}$

(8) $(-100)^0 = ?$

- (A) 1000 (B) 100
-
- (C) 10 (D) 1

Solution :

 $\therefore$ (D) 1

(9) $(2x^{-1})^{-2} = ?$

- (A) 1 (B) 0 (C)
- $4x^6$
- (D)
- $\frac{x^2}{4}$

Solution :

$$\begin{aligned} \Rightarrow (2)^{-2} \times x^{-1 \times -2} &= \frac{1}{2^2} \times x^2 \\ &= \frac{x^2}{4} \end{aligned}$$

 $\therefore$ (D) $\frac{x^2}{4}$

(10) $12x^4 \div 3x^6$

- (A)
- $4x^{-2}$
- (B)
- $4x^2$
-
- (C)
- $2x^2$
- (D)
- $-4x^2$

Solution :

$$\begin{aligned} \Rightarrow \frac{\cancel{12}^4 x^4}{\cancel{3} x^6} &= 4 \times x^{4-6} \\ &= 4x^{-2} \end{aligned}$$

 $\therefore$ (A) $4x^{-2}$

Type # 2

□ સમજૂતી :

$$\Rightarrow a^m \times a^n = a^{m+n}$$

$$\Rightarrow a^m \div a^n = a^{m-n} \quad 1^0 = 1$$

$$\Rightarrow (a^m)^n = a^{m \times n} \quad a^0 = 1$$

$$\Rightarrow a^{-2} = \frac{1}{a^2} \quad a^1 = a$$

$$\Rightarrow \frac{1}{a^{-3}} = a^3 \quad a^{-1} = \frac{1}{a}$$

(11) જો $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3}$ હોય, તો x ની કિંમત શોધો.

(A) $\frac{3}{2}$

(B) 3

(C) 2

(D) $\frac{2}{3}$

Solution :

$$\Rightarrow \left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3}$$

$$(\because -(x-3) = -x+3)$$

$$\Rightarrow \left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{3-x}$$

$$\therefore x-1 = 3-x$$

$$x+x = 3+1$$

$$2x = 4$$

$$x = \frac{4}{2}$$

$$x = 2$$

$$\therefore (C) 2$$

(12) જો $x = 5 + 2\sqrt{6}$ હોય, તો $\frac{x-1}{\sqrt{x}} = ?$

(A) $\sqrt{2}$

(B) $2\sqrt{2}$

(C) $\sqrt{3}$

(D) $2\sqrt{3}$

Solution :

$$\Rightarrow x = 5 + 2\sqrt{6}$$

$$x = 2 + 3 + 2\sqrt{6}$$

$$x = (\sqrt{2} + \sqrt{3})^2$$

$$\sqrt{x} = \sqrt{2} + \sqrt{3}$$

$$\frac{x-1}{\sqrt{x}} = \frac{5+2\sqrt{6}-1}{\sqrt{2}+\sqrt{3}}$$

$$= \frac{4+2\sqrt{6}}{\sqrt{2}+\sqrt{3}}$$

$$= \frac{2\sqrt{2}(\sqrt{2}+\sqrt{3})}{\sqrt{2}+\sqrt{3}}$$

$$= 2\sqrt{2}$$

$$\therefore (B) 2\sqrt{2}$$

(13) $\frac{(243)^{0.13} \times (243)^{0.07}}{7^{0.25} \times (49)^{0.075} \times (343)^{0.2}} = ?$

(A) $\frac{3}{7}$

(B) $\frac{5}{7}$

(C) $\frac{7}{3}$

(D) $\frac{7}{5}$

Solution :

$$= \frac{(3^5)^{0.13} \times (3^5)^{0.07}}{7^{0.25} \times (7^2)^{0.075} \times (7^3)^{0.2}}$$

$$= \frac{3^{0.65} \times 3^{0.35}}{7^{0.25} \times 7^{0.15} \times 7^{0.6}}$$

$$= \frac{3^{0.65+0.35}}{7^{0.25+0.15+0.6}}$$

$$= \frac{3^1}{7^1}$$

$$= \frac{3}{7}$$

$$\therefore (A) \frac{3}{7}$$

(14) $\frac{9^n \times 3^5 \times (27)^3}{3 \times (81)^4} = 3^3$ હોય તો $n = ?$

(A) 0

(B) 2

(C) 3

(D) 4

Solution :

$$= \frac{(3^2)^n \times 3^5 \times (3^3)^3}{3^1 \times (3^4)^4} = 3^3$$

$$\frac{3^{2n} \times 3^5 \times 3^9}{3^1 \times 3^{16}} = 3^3$$

$$3^{2n+14-17} = 3^3$$

$$3^{2n-3} = 3^3$$

$$2n-3 = 3$$

$$2n = 3+3$$

$$2n = 6$$

$$n = \frac{6}{2}$$

$$n = 3$$

$$\therefore (C) 3$$

(15) $5^x \times 25^{x-1} \div (5^{x-1} \times 25^{x-1}) = ?$

(A) 2 (B) 5 (C) 7 (D) 9

Solution :

$$\begin{aligned} \Rightarrow \frac{5^x \times (5^2)^{x-1}}{5^{x-1} \times (5^2)^{x-1}} &= \frac{5^x \times 5^{2x-2}}{5^{x-1} \times 5^{2x-2}} \\ &= \frac{5^{x+2x-2}}{5^{x-1+2x-2}} \\ &= \frac{5^{3x-2}}{5^{3x-3}} \\ &= 5^{\cancel{3x-2}-\cancel{3x}+3} \\ &= 5^1 \\ &= 5 \end{aligned}$$

$\therefore$ (B) 5

(16) જો M અને n પૂર્ણ સંખ્યાઓ છે. જો $M^n = 121$ હોય, તો $(M-1)^{n+1} = ?$

(A) 100 (B) 1000
(C) 10000 (D) 10

Solution :

$$\begin{aligned} \Rightarrow 11^2 &= 121 \quad (\because M^n = 121) \\ n &= 2, M = 11 \\ \Rightarrow (M-1)^{n+1} &= (11-1)^{2+1} \\ &= (10)^3 \\ &= 1000 \end{aligned}$$

$\therefore$ (B) 1000

(17) $\sqrt{\frac{(0.1)^2 + (0.01)^2 + (0.009)^2}{(0.01)^2 + (0.001)^2 + (0.0009)^2}} = ?$

(A) 10^2 (B) 10
(C) 0.1 (D) 0.01

Solution :

$$\begin{aligned} \Rightarrow \sqrt{\frac{(0.1)^2 + (0.01)^2 + (0.009)^2}{(0.01)^2 + (0.001)^2 + (0.0009)^2}} \\ &= \sqrt{\frac{(0.1)^2 + (0.01)^2 + (0.009)^2}{\frac{(0.1)^2}{10^2} + \frac{(0.01)^2}{10^2} + \frac{(0.009)^2}{10^2}}} \\ &= \sqrt{\frac{\cancel{(0.1)^2} + \cancel{(0.01)^2} + (0.009)^2}{\frac{1}{10^2} [\cancel{(0.1)^2} + \cancel{(0.01)^2} + (0.009)^2]}} \end{aligned}$$

$$\begin{aligned} &= \sqrt{\frac{1}{\frac{1}{10^2}}} \\ &= \sqrt{10^2} \\ &= \sqrt{100} \\ &= 10 \end{aligned}$$

$\therefore$ (B) 10

(18) $\sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{(0.003)^2 + (0.021)^2 + (0.0065)^2}} = ?$

(A) 0.1 (B) 10
(C) 10^2 (D) 10^3

Solution :

$$\begin{aligned} \Rightarrow \sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{(0.003)^2 + (0.021)^2 + (0.0065)^2}} \\ &= \sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{\frac{(0.03)^2}{10^2} + \frac{(0.21)^2}{10^2} + \frac{(0.065)^2}{10^2}}} \\ &= \sqrt{\frac{1(\cancel{0.03})^2 + \cancel{(0.21)^2} + \cancel{(0.065)^2}}{\frac{1}{10^2} [\cancel{(0.03)^2} + \cancel{(0.21)^2} + \cancel{(0.065)^2}]}} \\ &= \sqrt{\frac{1}{\frac{1}{10^2}}} \\ &= \sqrt{10^2} \\ &= \sqrt{100} \\ &= 10 \end{aligned}$$

$\therefore$ (B) 10

(19) જો 175616 નું ઘનમૂળ 56 છે તો,

$$\sqrt[3]{175.616} + \sqrt[3]{0.175616} + \sqrt[3]{0.000175616} = ?$$

(A) 0.168 (B) 62.16
(C) 6.216 (D) 6.116

Solution :

$$\Rightarrow 5.6 + 0.56 + 0.056 = 6.216$$

$\therefore$ (C) 6.216

Type # 3

(20) $2^{60}, 3^{48}, 4^{36}, 5^{24}$ માં સૌથી મોટી સંખ્યા કઈ છે ?

- (A)
- 2^{60}
- (B)
- 3^{48}
- (C)
- 4^{36}
- (D)
- 5^{24}

Solution :

$$\Rightarrow \begin{array}{cccc} 60 & 48 & 36 & 24 \text{ (ઘાતનો ગુ.સા.અ.)} \\ (12 \times 5) & (12 \times 4) & (12 \times 3) & (12 \times 2) \\ (2^5)^{12} & (3^4)^{12} & (4^3)^{12} & (5^2)^{12} \\ (32)^{12} & (81)^{12} & (64)^{12} & (25)^{12} \end{array}$$

$$\Rightarrow (81)^{12} = 3^{48}$$

$$\therefore \text{ (B) } 3^{48}$$

(21) $0.16, \sqrt{0.16}, (0.16)^2, 0.04$ માં સૌથી મોટી સંખ્યા કઈ છે ?

- (A) 0.16 (B)
- $\sqrt{0.16}$
-
- (C)
- $(0.16)^2$
- (D) 0.04

Solution :

$$\Rightarrow 0.16, \sqrt{0.16}, (0.16)^2, 0.04$$

$$\Rightarrow 0.16, 0.4, 0.0256, 0.04$$

$$\therefore \text{ (B) } \sqrt{0.16}$$

(22) $(14)^{\frac{1}{3}}, (12)^{\frac{1}{2}}, (16)^{\frac{1}{6}}$ અને $(25)^{\frac{1}{12}}$ માંથી સૌથી નાની સંખ્યા કઈ છે ?

- (A)
- $(14)^{\frac{1}{3}}$
- (B)
- $(25)^{\frac{1}{12}}$
-
- (C)
- $(16)^{\frac{1}{6}}$
- (D)
- $(12)^{\frac{1}{2}}$

Solution :

$$\Rightarrow = (14)^{\frac{1}{3} \times \frac{4}{4}}, (12)^{\frac{1}{2} \times \frac{6}{6}}, (16)^{\frac{1}{6} \times \frac{2}{2}}, (25)^{\frac{1}{12} \times \frac{1}{12}}$$

$$3, 2, 6, 12 \rightarrow 12 \text{ લ.સા.અ.}$$

$$\Rightarrow 14^4, 12^6, 16^2, \boxed{25^1}$$

$$\Rightarrow 196 \times 196, 144 \times 144 \times 144, 256 \times 256, \boxed{25}$$

$$\therefore \text{ (B) } (25)^{\frac{1}{12}}$$

(23) $\sqrt{5}, \sqrt[3]{7}, \sqrt[4]{13}$ આમાંથી મોટી સંખ્યા કઈ ?

- (A)
- $\sqrt{5}$
- (B)
- $3\sqrt{7}$
-
- (C)
- $4\sqrt{13}$
- (D) બધી સમાન

Solution :

$$\Rightarrow (5)^{\frac{1}{2} \times \frac{6}{6}}, (7)^{\frac{1}{3} \times \frac{4}{4}}, (13)^{\frac{1}{4} \times \frac{3}{3}}$$

$$\Rightarrow (5)^{\frac{6}{12}}, (7)^{\frac{4}{12}}, (13)^{\frac{3}{12}}$$

$$\therefore \text{ (A) } \sqrt{5}$$

(24) $\sqrt[3]{6}, \sqrt[2]{5}$ અને $\sqrt[6]{12}$ આમાંથી સૌથી મોટી સંખ્યા કઈ છે ?

- (A)
- $\sqrt[3]{6}$
- (B)
- $\sqrt[2]{5}$
-
- (C)
- $\sqrt[6]{12}$
- (D) બધી સમાન

Solution :

$$\Rightarrow (6)^{\frac{1}{3}}, (5)^{\frac{1}{2}}, (12)^{\frac{1}{6}} \text{ લ.સા.અ. 6}$$

$$= (6)^{\frac{1}{3} \times \frac{2}{2}}, (5)^{\frac{1}{2} \times \frac{3}{3}}, (12)^{\frac{1}{6} \times \frac{1}{1}}$$

$$= 6^2, 5^3, 12$$

$$= 36, 125, 12$$

$$\therefore \text{ (B) } \sqrt[2]{5}$$

(25) $\sqrt[3]{8}, \sqrt[4]{13}, \sqrt[5]{16}, \sqrt[10]{41}$ આમાંથી સૌથી મોટી સંખ્યા કઈ છે ?

- (A)
- $\sqrt[4]{13}$
- (B)
- $\sqrt[5]{16}$
-
- (C)
- $\sqrt[10]{41}$
- (D)
- $\sqrt[3]{8}$

Solution :

$$\Rightarrow \frac{1}{8^3}, \frac{1}{13^4}, \frac{1}{16^5}, \frac{1}{41^{10}} \quad (3, 4, 5, 10 \rightarrow 60)$$

$$\Rightarrow \frac{1}{8^{\frac{1}{3} \times \frac{20}{20}}}, \frac{1}{13^{\frac{1}{4} \times \frac{15}{15}}}, \frac{1}{16^{\frac{1}{5} \times \frac{12}{12}}}, \frac{1}{41^{\frac{1}{10} \times \frac{6}{6}}}$$

$$\Rightarrow 8^{20}, 13^{15}, 16^{12}, 41^6$$

$$\Rightarrow (8^4)^5, (13^3)^5, (16^2)^6, (41)^6$$

$$\therefore \text{ (D) } \sqrt[3]{8}$$

(26) $3^{34}, 2^{51}, 7^{17}$ આ સંખ્યાઓને ઉતરતા ક્રમમાં ગોઠવો.

- (A)
- $3^{34} > 2^{51} > 7^{17}$
- (B)
- $7^{17} < 2^{51} < 3^{34}$
-
- (C)
- $3^{34} > 7^{17} > 2^{51}$
- (D)
- $2^{51} > 3^{34} > 7^{17}$

Solution :

$$\Rightarrow 34, 51, 17$$

$$= (3^2)^{17}, (2^3)^{17}, (7)^{17}$$

$$= (9)^{17}, (8)^{17}, (7)^{17}$$

$$= 7^{17} < 8^{17} < 9^{17}$$

$$= 7^{17} < 2^{51} < 3^{34} \text{ (ઉતરતા ક્રમમાં ગોઠવતા)}$$

$$\therefore \text{ (A) } 3^{34} > 2^{51} > 7^{17}$$

$$(33) \sqrt{12\sqrt{12\sqrt{12\sqrt{12\sqrt{12}}}}}=?$$

- (A) $12^{\frac{32}{31}}$ (B) $12^{\frac{64}{63}}$
 (C) $12^{\frac{31}{32}}$ (D) $12^{\frac{63}{64}}$

Solution :

$$\Rightarrow 2^5 = 32$$

$$\Rightarrow 12^{\frac{31}{32}}$$

$$\therefore (C) 12^{\frac{31}{32}}$$

$$(34) \sqrt{72 + \sqrt{72 + \sqrt{72 + \sqrt{72 + \dots}}}}=?$$

- (A) 4 (B) 5 (C) 8 (D) 9

Solution :

$$\Rightarrow a = 72$$

$$\Rightarrow \frac{\sqrt{4a+1}+1}{2} = \frac{\sqrt{4 \times 72 + 1} + 1}{2}$$

$$= \frac{\sqrt{288+1}+1}{2}$$

$$= \frac{\sqrt{289}+1}{2}$$

$$= \frac{17+1}{2}$$

$$= \frac{18}{2}$$

$$= 9$$

$$\therefore (D) 9$$

$$(35) \sqrt{72 - \sqrt{72 - \sqrt{72 - \sqrt{72 - \dots}}}}=?$$

- (A) 4 (B) 5 (C) 8 (D) 9

Solution :

$$\Rightarrow (a = 72)$$

$$\Rightarrow \frac{\sqrt{4a+1}-1}{2} = \frac{\sqrt{288+1}-1}{2}$$

$$= \frac{\sqrt{289}-1}{2}$$

$$= \frac{17-1}{2}$$

$$= 8$$

$$= \frac{16}{2}$$

$$= 8$$

$$\therefore (C) 8$$

$$(36) \sqrt{5 + \sqrt{5 + \sqrt{5 + \dots}}}=?$$

- (A) $\frac{\sqrt{21}+1}{2}$ (B) $\frac{\sqrt{21}-1}{2}$ (C) $\frac{\sqrt{21}+3}{2}$ (D) $\frac{\sqrt{21}-3}{2}$

Solution :

$$\Rightarrow (a = 5)$$

$$\Rightarrow \frac{\sqrt{4a+1}+1}{2} = \frac{\sqrt{20+1}+1}{2} = \frac{\sqrt{21}+1}{2}$$

$$\therefore (A) \frac{\sqrt{21}+1}{2}$$

$$(37) \sqrt{1 - \sqrt{1 - \sqrt{1 - \dots}}}=?$$

- (A) $\frac{\sqrt{5}+1}{2}$ (B) $\frac{\sqrt{5}-1}{2}$ (C) $\frac{\sqrt{5}+3}{2}$ (D) $\frac{\sqrt{5}-3}{2}$

Solution :

$$\Rightarrow (a = 1)$$

$$\Rightarrow \frac{\sqrt{4a+1}-1}{2} = \frac{\sqrt{4+1}-1}{2} = \frac{\sqrt{5}-1}{2}$$

$$\therefore (B) \frac{\sqrt{5}-1}{2}$$

$$(38) \sqrt{35 + 2\sqrt{35 + 2\sqrt{35 + 2\sqrt{35 + \dots}}}}=?$$

- (A) 6 (B) 7 (C) 5 (D) 6.4

Solution :

$$\begin{matrix} 35 \\ \wedge \\ 5 \times 7 \end{matrix}$$

$$\Rightarrow a = 35, b = 2 \text{ (દરેક 7-5 = 2)}$$

$$\Rightarrow \frac{\sqrt{4a+b^2}+b}{2} = \frac{\sqrt{140+4}+2}{2}$$

$$= \frac{\sqrt{144}+2}{2}$$

$$= \frac{12+2}{2}$$

$$= \frac{14}{2}$$

$$= 7$$

$$\therefore (B) 7$$

(39) $\sqrt{35-2\sqrt{35-2\sqrt{35-2\sqrt{35-\dots\infty}}}}=?$
 (A) 6 (B) 7 (C) 5 (D) 6.4

Solution :

$\Rightarrow a = 35, b = 2$

$$\begin{aligned}\Rightarrow \frac{\sqrt{4a+b^2}-b}{2} &= \frac{\sqrt{140+4}-2}{2} \\ &= \frac{\sqrt{144}-2}{2} \\ &= \frac{12-2}{2} \\ &= \frac{10}{2} \\ &= 5\end{aligned}$$

$\therefore$ (C) 5

(40) $\sqrt{154-3\sqrt{154-3\sqrt{154-3\sqrt{154-\dots\infty}}}}=?$
 (A) 3 (B) 14 (C) 11 (D) 9

Solution :

$\Rightarrow a = 154, b = 3$

$$\begin{aligned}\Rightarrow \frac{\sqrt{4a+b^2}-b}{2} &= \frac{\sqrt{616+9}-3}{2} \\ &= \frac{\sqrt{625}-3}{2} \\ &= \frac{25-3}{2} \\ &= \frac{22}{2} \\ &= 11\end{aligned}$$

$\therefore$ (C) 11

(41) જો $A = \sqrt{11+3\sqrt{11+3\sqrt{11+3\sqrt{11+\dots\infty}}}}$
 અને $B = \sqrt{11-3\sqrt{11-3\sqrt{11-3\sqrt{11-\dots\infty}}}}$
 હોય, તો $A+B=?$

(A) $\sqrt{47}$ (B) $\sqrt{65}$ (C) $\sqrt{41}$ (D) $\sqrt{53}$

Solution :

$\Rightarrow A = \frac{\sqrt{4a+b^2}+b}{2}, B = \frac{\sqrt{4a+b^2}-b}{2}$

$\Rightarrow A-B=b$

$A \times B = a$

$\Rightarrow A-B=b=3$

$A \times B = a = 11$

$\Rightarrow (A+B)^2 = (A-B)^2 + 4AB$
 $= (3)^2 + 4(11)$
 $= 9 + 44$

$(A+B)^2 = 53$

$A+B = \sqrt{53}$

$\therefore$ (D) $\sqrt{53}$

(42) $\sqrt{7+2\sqrt{7-2\sqrt{7+2\sqrt{7-2\sqrt{7+\dots\infty}}}}=?$
 (A) $\sqrt{51}$ (B) 4
 (C) $\frac{3+\sqrt{15}}{2}$ (D) 3

Solution :

$\Rightarrow a = 7, b = 2$

$$\begin{aligned}\Rightarrow \frac{\sqrt{4a-3b^2}+b}{2} &= \frac{\sqrt{28-12}+2}{2} \\ &= \frac{\sqrt{16}+2}{2} \\ &= \frac{4+2}{2} \\ &= \frac{6}{2} \\ &= 3\end{aligned}$$

$\therefore$ (D) 3

(43) $\sqrt{13-3\sqrt{13+3\sqrt{13-3\sqrt{13+\dots\infty}}}}=?$
 (A) $\sqrt{51}$ (B) 1
 (C) $\frac{3+\sqrt{15}}{2}$ (D) 3

Solution :

$\Rightarrow a = 13, b = 3$

$$\Rightarrow \frac{\sqrt{4a-3b^2}-b}{2} = \frac{\sqrt{52-27}-3}{2}$$

$$= \frac{\sqrt{25}-3}{2}$$

$$= \frac{5-3}{2}$$

$$= \frac{1}{1}$$

$$= 1$$

∴ (B) 1

(44) $\sqrt{6-\sqrt{6+\sqrt{6-\sqrt{6+\dots\infty}}}}=?$

(A) $\frac{\sqrt{21}+1}{2}$ (B) $\frac{\sqrt{21}-1}{2}$
 (C) 3 (D) 4

Solution :

$$\Rightarrow a = 6, b = 1$$

$$\Rightarrow \frac{\sqrt{4a-3}-1}{2} = \frac{\sqrt{24-3}-1}{2} = \frac{\sqrt{21}-1}{2}$$

∴ (B) $\frac{\sqrt{21}-1}{2}$

(45) $\sqrt{13+\sqrt{13-\sqrt{13+\sqrt{13-\dots\infty}}}}=?$

(A) $\frac{\sqrt{51}+1}{2}$ (B) $\frac{\sqrt{51}-1}{2}$
 (C) 3 (D) 4

Solution :

$$\Rightarrow a = 13$$

$$\Rightarrow \frac{\sqrt{4a-3}+1}{2} = \frac{\sqrt{52-3}+1}{2}$$

$$= \frac{\sqrt{49}+1}{2}$$

$$= \frac{7+1}{2}$$

$$= \frac{8}{2}$$

$$= 4$$

∴ (D) 4

Type # 5

(46) નીચેનામાંથી કઈ સંખ્યા મોટી છે ?

- (A) $\sqrt{7} + \sqrt{9}$ (B) $\sqrt{6} + \sqrt{10}$
 (C) $\sqrt{5} + \sqrt{11}$ (D) $\sqrt{4} + \sqrt{12}$

Solution :

$$\therefore (A) \sqrt{7} + \sqrt{9}$$

⇒ સરવાળો સરખો હોય તો અને '+' હોય તો જેનો ગુણાકાર મોટો તે સંખ્યા મોટી બને.

(47) $\sqrt{11} + \sqrt{5}, \sqrt{14} + \sqrt{2}, \sqrt{8} + \sqrt{8}$ માંથી કઈ સંખ્યા મોટી છે ?

- (A) $\sqrt{11} + \sqrt{5}$ (B) $\sqrt{14} + \sqrt{2}$
 (C) $\sqrt{8} + \sqrt{8}$ (D) બધી સમાન

Solution :

$$\Rightarrow \sqrt{11} + \sqrt{5}, \sqrt{14} + \sqrt{2}, \sqrt{8} + \sqrt{8}$$

16	16	16	સરવાળો
55	28	64	ગુણાકાર

$$\therefore (C) \sqrt{8} + \sqrt{8}$$

(48) $\sqrt{10} + \sqrt{4}, \sqrt{11} + \sqrt{3}, \sqrt{7} + \sqrt{7}$ માંથી કઈ સંખ્યા નાની છે ?

- (A) $\sqrt{10} + \sqrt{4}$ (B) $\sqrt{11} + \sqrt{3}$
 (C) $\sqrt{7} + \sqrt{7}$ (D) બધી સમાન

Solution :

$$\Rightarrow \sqrt{10} + \sqrt{4}, \sqrt{11} + \sqrt{3}, \sqrt{7} + \sqrt{7}$$

14	14	14	સરવાળો
40	33	49	ગુણાકાર

$$\therefore (B) \sqrt{11} + \sqrt{3}$$

(49) $\sqrt{7}-\sqrt{5}, \sqrt{9}-\sqrt{7}, \sqrt{5}-\sqrt{3}, \sqrt{11}-\sqrt{9}$ આમાંથી કઈ સંખ્યા મોટી છે ?

- (A) $\sqrt{7}-\sqrt{5}$ (B) $\sqrt{9}-\sqrt{7}$
 (C) $\sqrt{5}-\sqrt{3}$ (D) $\sqrt{11}-\sqrt{9}$

Solution :

35	63	15	99
$\sqrt{7}-\sqrt{5}$	$\sqrt{9}-\sqrt{7}$	$\sqrt{5}-\sqrt{3}$	$\sqrt{11}-\sqrt{9}$
2	2	2	2

$$\therefore (C) \sqrt{5}-\sqrt{3}$$

⇒ તફાવત સરખો હોય તો જેનો ગુણાકાર નાનો તે સંખ્યા મોટી અને જેનો ગુણાકાર મોટો તે સંખ્યા નાની.

- (50) $\sqrt{8} + \sqrt{3}, \sqrt{6} + \sqrt{4}, \sqrt{12} + \sqrt{2}$ અને $\sqrt{24} + \sqrt{1}$ માંથી સૌથી મોટી સંખ્યા કઈ છે ?
- (A) $\sqrt{8} + \sqrt{3}$ (B) $\sqrt{6} + \sqrt{4}$
(C) $\sqrt{12} + \sqrt{2}$ (D) $\sqrt{24} + \sqrt{1}$

Solution :

$$\Rightarrow \frac{11}{24} + \frac{10}{24}, \frac{14}{24} + \frac{25}{24}, \frac{14}{24} + \frac{25}{24} \text{ ગુણકાર}$$

$\Rightarrow$ '+' $\rightarrow$ વધારે
 $\rightarrow$ સંખ્યા મોટી

$$\therefore \text{(D) } \sqrt{24} + \sqrt{1}$$

- (51) $\sqrt{5}-\sqrt{4}, \sqrt{10}-\sqrt{2}, \sqrt{20}-\sqrt{1}$ આમાંથી સૌથી નાની સંખ્યા કઈ છે ?
- (A) $\sqrt{5}-\sqrt{4}$ (B) $\sqrt{10}-\sqrt{2}$
(C) $\sqrt{20}-\sqrt{1}$ (D) બધી સમાન

Solution :

$$\Rightarrow \frac{1}{20} - \frac{8}{20}, \frac{8}{20} - \frac{19}{20}, \frac{19}{20} - \frac{1}{20}$$

$$\therefore \text{(A) } \sqrt{5}-\sqrt{4}$$

- (52) $\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{4}+\sqrt{3}} + \dots + \frac{1}{\sqrt{100}+\sqrt{99}} = ?$
- (A) 1 (B) 9
(C) $\sqrt{99}$ (D) $\sqrt{99}-1$

Solution :

$$\Rightarrow \frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{4}+\sqrt{3}} + \dots + \frac{1}{\sqrt{100}+\sqrt{99}} = ?$$

$$= \cancel{\sqrt{2}} - 1 + \sqrt{3} - \cancel{\sqrt{2}} + \dots + \sqrt{100} - \cancel{\sqrt{99}} - 1 + \sqrt{100}$$

$$= -1 + \sqrt{100}$$

$$= -1 + 10$$

$$= 9$$

$$\therefore \text{(B) } 9$$

- (53) $\frac{1}{\sqrt{3}+\sqrt{4}} + \frac{1}{\sqrt{4}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{6}} + \frac{1}{\sqrt{6}+\sqrt{7}} = ?$
- (A) $\sqrt{3}-\sqrt{7}$ (B) $\sqrt{6}-\sqrt{5}$
(C) $\sqrt{7}-\sqrt{3}$ (D) $\sqrt{8}-\sqrt{5}$

Solution :

$$\Rightarrow \frac{1}{\sqrt{3}+\sqrt{4}} + \frac{1}{\sqrt{4}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{6}} + \frac{1}{\sqrt{6}+\sqrt{7}}$$

$$= \frac{1}{\sqrt{4}+\sqrt{3}} + \frac{1}{\sqrt{5}+\sqrt{4}} + \frac{1}{\sqrt{6}+\sqrt{5}} + \frac{1}{\sqrt{7}+\sqrt{6}}$$

$$= \cancel{\sqrt{4}} - \sqrt{3} + \cancel{\sqrt{5}} - \cancel{\sqrt{4}} + \cancel{\sqrt{6}} - \cancel{\sqrt{5}} + \sqrt{7} - \cancel{\sqrt{6}}$$

$$= -\sqrt{3} + \sqrt{7}$$

$$= \sqrt{7} - \sqrt{3}$$

$$\therefore \text{(C) } \sqrt{7} - \sqrt{3}$$

Type # 6

- (54) $49 \times 64 = (?)^2$, તો =

- (A) 28 (B) 63
(C) 56 (D) 42

Solution :

$$\Rightarrow x^2 = 49 \times 64$$

$$= 7 \times 8$$

$$= 56$$

$$\therefore \text{(C) } 56$$

- (55) $(2^3)^x = 64$ હોય, તો $x = ?$

- (A) 8 (B) 16
(C) 2 (D) 4

Solution :

$$\Rightarrow 2^{3 \times x} = 2^6$$

$$2^{3x} = 2^6$$

$$3x = 6$$

$$x = \frac{6}{3}$$

$$x = 2$$

$$\therefore \text{(C) } 2$$

- (56) $(0.6)^4 = \dots\dots$

- (A) 12.96 (B) 1.296
(C) 0.1296 (D) 129.6

Solution :

$$\Rightarrow \left(\frac{6}{10}\right)^4 = \frac{6^4}{10^4}$$

$$= \frac{1296}{10000}$$

$$= 0.1296$$

$$\therefore \text{(C) } 0.1296$$

(57) $\sqrt{225} - 5\sqrt[3]{125} = \dots$

- (A) 10 (B) (-10) (C) 0 (D) 50

Solution :

$$= 15 - 5 \times 5$$

$$= 15 - 25$$

$$= (-10)$$

$$\therefore \text{(B) } (-10)$$

(58) $4 \times 16 \times 64 = (b)^3$ નું $b = \dots$

- (A) 8 (B) 12 (C) 14 (D) 16

Solution :

$$\Rightarrow b^3 = 4 \times 16 \times 64$$

$$\Rightarrow b^3 = 64 \times 64$$

$$b = \sqrt[3]{64 \times 64}$$

$$b = 4 \times 4$$

$$b = 16$$

$$\therefore \text{(D) } 16$$

(59) $64 \times 81 = (?)^2$? =

- (A) 72 (B) 25 (C) 20 (D) 15

Solution :

$$\Rightarrow x^2 = 64 \times 81$$

$$x = \sqrt{64 \times 81}$$

$$x = 8 \times 9$$

$$x = 72$$

$$\therefore \text{(A) } 72$$

(60) $1^{100} + 100^0 = ?$

- (A) 1 (B) 2 (C) 200 (D) 101

Solution :

$$\Rightarrow 1 + 1 = 2$$

$$\therefore \text{(B) } 2$$

(61) $\sqrt{16 \times 256} = (\dots)^2$

- (A) 16 (B) 8 (C) 7 (D) 9

Solution :

$$\Rightarrow x^2 = \sqrt{16 \times 256}$$

$$x^2 = 4 \times 16$$

$$x^2 = 64$$

$$x = \sqrt{64}$$

$$x = 8$$

$$\therefore \text{(B) } 8$$

(62) $\sqrt{3 + \sqrt[3]{x}} = 3$ નો ઉકેલ છે.

- (A) 27 (B) 126 (C) 216 (D) 621

Solution :

$$\Rightarrow \sqrt{3 + \sqrt[3]{x}} = 3$$

$$3 + \sqrt[3]{x} = 9$$

$$\sqrt[3]{x} = 9 - 3$$

$$\sqrt[3]{x} = 6$$

$$x = (6)^3$$

$$x = 216$$

$$\therefore \text{(C) } 216$$

(63) $\sqrt[3]{x^{12}} \times \sqrt[2]{x^6} = \dots$

- (A)
- x^5
- (B)
- x^7
- (C)
- x^{18}
- (D)
- x^9

Solution :

$$\Rightarrow \sqrt[3]{x^{12}} \times \sqrt[2]{x^6} = x^{12 \times \frac{1}{3}} \times x^{6 \times \frac{1}{2}}$$

$$= x^4 \times x^3$$

$$= x^{4+3}$$

$$= x^7$$

$$\therefore \text{(B) } x^7$$

(64) $(3 \times 6)^3 \div (3^3 \times 6^2)$ ની કિંમત શું થાય ?

- (A) 12 (B) 3 (C) 9 (D) 6

Solution :

$$\Rightarrow \frac{(3 \times 6)^3}{3^3 \times 6^2} = \frac{\cancel{3}^3 \times \cancel{6}^3 \times 6^3}{\cancel{3}^3 \times \cancel{6}^2 \times 6^2}$$

$$= \frac{6^3}{6^2}$$

$$= 6^{3-2}$$

$$= 6^1$$

$$\therefore \text{(D) } 6$$

(65) $(a - b)^2 = \dots$

(A) $a^2 - b^2$

(B) $a^2 + 2ab + b^2$

(C) $a^2 + b^2$

(D) $a^2 - 2ab + b^2$

Solution :

$$\therefore \text{(D) } a^2 - 2ab + b^2$$

(66) $\frac{36y^6 \times 21y^3}{28y^4} = \underline{\hspace{2cm}}$

- (A) $9y^4$ (B) $21y^3$ (C) $27y^5$ (D) $21y^2$

Solution :

$$\Rightarrow \frac{36y^6 \times 21y^3}{28y^4} = \frac{\overset{9}{\cancel{36}} \times \overset{3}{\cancel{21}} \times y^{6+3-4}}{\cancel{28}} = 27 \times y^5 = 27y^5$$

$\therefore$ (C) $27y^5$

(67) $\frac{a^5 \times ab^6 \times a^2b}{a^3b^4} = \underline{\hspace{2cm}}$

- (A) a^3b^5 (B) a^2b^6 (C) a^3b^2 (D) a^5b^3

Solution :

$$\Rightarrow a^{5+1+2-3} \times b^{6+1-4} = a^{8-3} \times b^{7-4} = a^5 \times b^3$$

$\therefore$ (D) a^5b^3

(68) $(3x - 7y)^2 = \underline{\hspace{2cm}}$

- (A) $9x^2 - 49y^2$ (B) $9x^2 - 42xy + 49y^2$
(C) $9x^2 + 42xy - 49y^2$ (D) $6x^2 - 21xy + 14y^2$

Solution :

$$\Rightarrow (a - b)^2 = a^2 - 2ab + b^2$$

$$\Rightarrow 9x^2 - 42xy + 49y^2$$

$\therefore$ (B) $9x^2 - 42xy + 49y^2$

(69) $a^2 - 0.4a + 0.04 = \underline{\hspace{2cm}}$

- (A) $(a + 0.02)^2$ (B) $(a - 0.02)^2$
(C) $(a - 0.2)^2$ (D) $(a + 0.2)^2$

Solution :

$$\Rightarrow (a - 0.2)^2$$

$$\Rightarrow a^2 - 0.4a + 0.04$$

$\therefore$ (C) $(a - 0.2)^2$

(70) $(3 \times 6)^2 \div 3^2 = \underline{\hspace{2cm}}$

- (A) 63 (B) 81 (C) 36 (D) 18

Solution :

$$\Rightarrow \frac{(3 \times 6)^2}{3^2} = \frac{\cancel{3}^2 \times 6^2}{\cancel{3}^2} = 6^2 = 36$$

$\therefore$ (C) 36

(71) $\left(\frac{2}{3}\right)^7 \times \left(\frac{2}{3}\right)^3 \div \left(\frac{2}{3}\right)^6 = \underline{\hspace{2cm}}$

- (A) $\frac{2}{3}$ (B) $\frac{4}{9}$ (C) $\frac{8}{27}$ (D) $\frac{16}{81}$

Solution :

$$\Rightarrow \left(\frac{2}{3}\right)^{7+3-6} = \left(\frac{2}{3}\right)^4 = \frac{2^4}{3^4} = \frac{16}{81}$$

$\therefore$ (D) $\frac{16}{81}$

(72) $x^{10} \times x^{-7} \times x^0 = \underline{\hspace{2cm}}$

- (A) x^7 (B) x^0 (C) x^6 (D) x^3

Solution :

$$\Rightarrow x^{10} \times x^{-7} \times x^0 = x^{10+(-7)+0} = x^{10-7} = x^3$$

$\therefore$ (D) x^3

(73) $x^7 \div (x^4 \times x^2) = \underline{\hspace{2cm}}$

- (A) x^2 (B) x^0 (C) x (D) x^5

Solution :

$$\Rightarrow x^7 \div (x^4 \times x^2) = \frac{x^7}{x^4 \times x^2} = \frac{x^7}{x^{4+2}} = \frac{x^7}{x^6} = x^{7-6} = x^1 = x$$

$\therefore$ (C) x

(74) $\left(\frac{a^6}{a^2}\right)^3 = \underline{\hspace{2cm}}$

- (A) a^{12} (B) a^{10} (C) a^3 (D) a^6

Solution :

$$\Rightarrow \left(\frac{a^6}{a^2}\right)^3 = (a^{6-2})^3 = (a^4)^3 = a^{12}$$

$\therefore$ (A) a^{12}

(75) $(1)^{1000} - (1000)^0 = \underline{\hspace{2cm}}$

- (A) 999 (B) 0 (C) 1 (D) 1000

Solution :

$\Rightarrow 1 - 1 = 0$

$\therefore (B) 0$

(76) 5^n નો અંતિમ અંક છે.

- (A) 4 (B) 0 (C) 5 (D) 2

Solution :

$\Rightarrow 5^1 = 5$

$\Rightarrow 5^2 = 25$

$\therefore (C) 5$

(77) $(x+y)^2 - (x-y)^2 = \underline{\hspace{2cm}}$

- (A)
- xy
- (B)
- $4xy$
-
- (C)
- $2xy$
- (D)
- $4x^2y^2$

Solution :

$$\begin{aligned} \Rightarrow (x^2 + 2xy + y^2) - (x^2 - 2xy + y^2) \\ = x^2 + 2xy + y^2 - x^2 + 2xy - y^2 \\ = 2xy + 2xy \\ = 4xy \end{aligned}$$

$\therefore (B) 4xy$

(78) $3a^2 + 4b^2$ અને $-3a^2 + 4b^2$ નો ગુણાકાર મેળવો.

- (A)
- $9a^4 - 16b^4$
- (B)
- $16b^4 - 9a^4$
-
- (C)
- $9a^2 - 16b^2$
- (D)
- $9a^4 + 16b^4$

Solution :

$$\begin{aligned} \Rightarrow (3a^2 + 4b^2)(-3a^2 + 4b^2) \\ = -9a^4 + 12a^2b^2 - 12a^2b^2 + 16b^4 \\ = -9a^4 + 16b^4 \end{aligned}$$

$\therefore (B) 16b^4 - 9a^4$

(79) કિંમત શોધો : $(3^0 + 8^{-1}) \times 2^2$

- (A)
- $25/2$
- (B)
- (-28)
-
- (C)
- $1/2$
- (D)
- $9/2$

Solution :

$$\begin{aligned} \Rightarrow \left(1 + \frac{1}{8}\right) \times 4 &= \left(\frac{8+1}{8}\right) \times 4 \\ &= \left(\frac{9}{8}\right) \times 4 \end{aligned}$$

$$\begin{aligned} &= \frac{9}{8} \times \frac{4}{1} \\ &= \frac{9}{2} \end{aligned}$$

$\therefore (D) 9/2$

(80) $2 + \frac{2}{3}\sqrt{5}$ નું દ્વિપદી કરણીના સ્વરૂપમાં વર્ગમૂળ શું થાય ?

- (A)
- $\frac{\sqrt{15+3}}{9}$
- (B)
- $\frac{6+2\sqrt{5}}{3}$
-
- (C)
- $\frac{6+4\sqrt{3}}{9}$
- (D)
- $\frac{\sqrt{15+\sqrt{3}}}{3}$

Solution :

$$\begin{aligned} \Rightarrow \frac{6+2\sqrt{5}}{3} &= \frac{5+1+2\sqrt{5}}{3} \\ &= \frac{(\sqrt{5}+\sqrt{1})^2}{3} \\ &= \sqrt{\frac{(\sqrt{5}+\sqrt{1})^2}{3}} \\ &= \frac{\sqrt{5}+1}{\sqrt{3}} \\ &= \frac{\sqrt{5}+1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{\sqrt{15}+\sqrt{3}}{3} \end{aligned}$$

$\therefore (D) \frac{\sqrt{15}+\sqrt{3}}{3}$

(81) $\frac{5}{2}x - 7$ માં x નો ઘાત કયો છે.

- (A) 7 (B) 1 (C) 5 (D)
- $5/2$

Solution :

$\Rightarrow \therefore x$ ની ઘાત 1 છે.

$\therefore (B) 1$

(82) $(3^4)^2 \div 3^{2^3} = \dots\dots\dots$

- (A)
- 3^2
- (B)
- 3^{10}
- (C) 0 (D) 1

Solution :

$$\Rightarrow \frac{(3^4)^2}{3^{2^3}} = \frac{3^8}{3^8} = 3^{8-8} = 3^0$$

$\therefore (D) 1$